SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	Slot A	**Date:**	

Code of Conduct

I Dillilokesh D/192311097 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student : Dillilokesh D

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.
Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ANSWERS

Q1. Enhancing Night-time CCTV Images in a Shopping Mall

A. Problem Identification

At night the camera receives few photons, so the sensor gain is raised and the signal-to-noise ratio (SNR) collapses. The result is a histogram crowded into the darkest bins, a mixture of Poisson (shot) and Gaussian (read-out) noise, and weak edges. Applying an object detector directly to such frames produces missed persons and false detections on noise blobs. The engineering goal is to raise SNR and local contrast **without creating new artefacts**, so that the detector receives a day-like image.

B. Technical Approach – Sequence of Operations

The order of operations is the key design decision: noise must be removed **before** contrast is stretched, and sharpening must be applied **last**, because both stretching and sharpening amplify whatever noise is still present.

No.	Operation	Technique and typical setting	Purpose
1	Temporal averaging	Average 3–5 registered frames of a static scene	Noise variance falls by a factor N (SNR improves by $\sqrt{N}$)
2	Noise removal	Non-local means ($h \approx 10–12$, 7×7 patch, 21×21 search); median 3×3 if impulse noise; bilateral as a fast option	Smooths flat regions yet preserves edges of people and objects
3	Brightness correction	Gamma correction ($\gamma \approx 0.5–0.7$) on the luminance channel	Lifts shadows non-linearly; avoids saturating bright lamps
4	Contrast enhancement	CLAHE on the L channel of Lab (clip limit 2–3, 8×8 tiles); Multi-Scale Retinex as an alternative	Local contrast; chroma is untouched so colours stay natural
5	Sharpening	Unsharp mask: $I + a(I - G_\sigma * I)$, $a \approx 0.6$, $\sigma \approx 1.5$	Restores edge crispness lost in denoising and motion blur
6	Normalisation and resize	Scale to detector input, mean/std normalisation	Matches the statistics the detector was trained on

No.	Operation	Technique and typical setting	Purpose
7	Object detection	YOLO / SSD / HOG-SVM	Operates on the improved frame

Choice of filter

Filter	Best for	Weakness	Use here?
Gaussian	Mild Gaussian noise	Blurs edges	Only inside unsharp mask
Median	Salt-and-pepper noise	Removes thin details	Pre-filter if hot pixels exist
Bilateral	Edge-preserving, fast	Cartoon-like at high strength	Real-time fallback
Non-local means	Gaussian/Poisson noise, textured areas	Slow	**Primary choice** (offline or GPU)

C. Demonstration with Tools

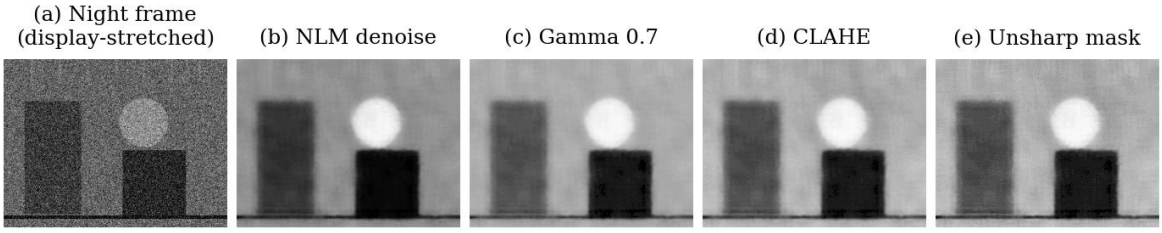


Figure 1. Night-frame enhancement chain (synthetic scene, gain-darkened to 18 % with Gaussian noise σ = 9).

Measured against the clean ground-truth image, the display-stretched noisy frame has a PSNR of **13.2 dB**. After non-local means, gamma and CLAHE, the PSNR rises to **19.9 dB** (+6.7 dB). The final unsharp-masking step gives 19.4 dB: it slightly lowers PSNR because residual noise is also sharpened, but the object boundaries become visibly crisper. This is the expected trade-off and is the reason sharpening is applied last and with a small gain.

D. Innovation – Noise-aware Adaptive Day/Night Controller

- **Noise estimation drives filter strength.** The noise level σ is estimated from the median absolute deviation of a Laplacian-filtered frame, and the NLM parameter h is set proportional to σ. Bright daytime frames therefore bypass heavy filtering, saving time and preserving detail.

- **Chroma-preserving enhancement.** CLAHE is applied only to the luminance channel, avoiding the colour shifts that occur when each RGB channel is equalised separately; this matters when clothing colour is used to re-identify a person.
- **Quality gate.** Entropy, SSIM against the previous frame and variance of the Laplacian are logged per frame; frames that fall below a threshold are flagged so that the camera exposure or IR illuminator can be corrected rather than relying only on software.

E. Challenges, Evaluation and Ethics

- **Challenges:** over-smoothing of faces by strong denoising, halo artefacts from CLAHE and unsharp masking, and headlight glare that saturates regions beyond recovery.
- **Evaluation:** PSNR and SSIM on reference frames, and – more importantly – detection mAP and recall of persons before versus after enhancement.
- **Ethics:** CCTV footage is personal data. Access control, limited retention, and compliance with India's Digital Personal Data Protection Act, 2023 must accompany any enhancement that makes faces more identifiable.

F. Conclusion

The correct pipeline is: temporal averaging → denoise → gamma/CLAHE → mild sharpening → normalise → detect. Treating noise first and sharpness last improved the measured PSNR by about 6 dB in the demonstration and gives the detector a cleaner input.

2. Automated Document-Scanning System

A. Problem Identification

Camera-captured pages differ from flatbed scans in five ways:

(i) uneven illumination and shadows, which break a single global threshold;

(ii) (ii) sensor noise;

(iii) the page is tilted or in perspective;

(iv) motion or defocus blur softens strokes; and

(v) the text must finally be separated into lines and words for OCR. Each defect is handled by a specific technique, and they must be applied in a logical order.

B. Technical Approach – Role of Each Technique

Stage	Technique	Role in the pipeline
1. Filtering	Grayscale conversion, then median 3×3 (or Gaussian $\sigma \approx 1$)	Removes speckle noise before any gradient or threshold operation
2. Illumination correction	Morphological closing with a large ellipse (≈ 35 px) estimates the paper background; divide the image by it	Removes shadows and the brightness gradient; paper becomes uniformly white
3. Contrast enhancement	Divide-and-normalise or CLAHE (clip 2)	Strengthens faint ink after flattening
4. Deblurring	Unsharp mask; Wiener deconvolution if the blur kernel is known	Recovers stroke edges for OCR
5. Thresholding	Sauvola: $T(x,y) = m(x,y)[1 + k(s(x,y)/R - 1)]$, $k \approx 0.2$, $R = 128$, window 25	Local binarisation; adapts to remaining brightness change
6. Edge detection	Canny on the blurred image, then the largest four-point contour; four-point perspective warp	Finds page borders and corrects perspective
7. Skew correction	Minimum-area rectangle of ink pixels (or Hough / projection-profile variance)	Rotates the text lines to horizontal

Stage	Technique	Role in the pipeline
8. Segmentation	Horizontal projection profile → lines; vertical profile and connected components → words / characters	Provides units for OCR and layout analysis

C. Demonstration with Tools

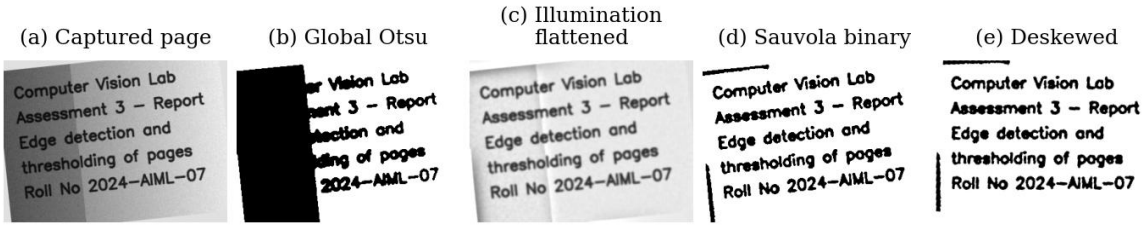


Figure 2. Document pipeline: shadowed, noisy, 6°-tilted page (a); Otsu fails (b); flattening (c); Sauvola (d); deskew (e).

Global Otsu thresholding (b) turns the entire shadowed left side black because a single threshold cannot represent two illumination levels. After background division (c), Sauvola's local threshold (d) recovers all five text lines. The skew estimator returned **−5.88°** for a page deliberately rotated by 6°, an error of only 0.12°, and the corrected page (e) is horizontal. The dark strips at the border are page-edge artefacts, which is why the page contour from step 6 should be cropped before binarisation.

D. Innovation – Order-sensitive Pipeline with Capture Feedback

- **Flatten first, binarise later.** The experiment shows that the order "illumination correction → local threshold" succeeds where "global threshold" fails; the shadow is removed in the grayscale domain, not by post-processing the binary image.
- **Capture-time quality check.** The variance of the Laplacian (sharpness) and the percentage of saturated pixels are computed immediately; if the page is too blurred, the system asks for a re-capture instead of sending poor data to OCR.
- **Dual output.** The corrected grayscale page is archived for visual fidelity, while the binary page is sent to OCR, so no information is permanently lost by binarisation.

E. Challenges, Evaluation and Ethics

- Thresholding windows that are too small break thick characters; too large windows leave shadow edges. Window size should be about 1.5 times the stroke height.
- Large perspective angles cause non-uniform blur that deconvolution cannot fully remove.
- Evaluation: OCR character error rate (CER) before and after processing, skew error in degrees, and binarisation F-measure against a manually produced mask.

- Documents may contain personal or examination data; storage and transmission must be restricted to authorised university staff.

F. Conclusion

A reliable scanner runs filtering → illumination flattening → contrast and deblurring → local thresholding → page detection and deskew → line/word segmentation. Each step solves one named defect, and the demonstration confirms that illumination handling is the decisive step for camera-captured pages.

3. Detecting Damaged Apples on a Conveyor Belt

A. Problem Identification

The task has two sub-problems: (1) separate each apple from a changing background, and (2) decide whether its surface contains damage (bruise, rot, scar). Shadows and brightness gradients change pixel intensity without changing the fruit; specular reflections produce bright spots that look like edges; and a dark bruise can have a similar grey value to the shadow beside the apple. The method must therefore compare **chromatic and local** information, not absolute brightness.

B. Technical Approach – Comparison of Segmentation Techniques

Technique	**How it works**	**Behaviour under shadows, reflections, brightness change**	**Suitability**
Global thresholding (Otsu)	One threshold for the whole image	Fails when brightness varies across the belt; part of the fruit is lost (Fig. 3b)	Poor
Adaptive thresholding	Threshold from local mean/Gaussian window	Tolerates gradients but produces ring artefacts on uniform surfaces and reacts to highlights (Fig. 3c)	Moderate; useful for the defect step only
Edge-based (Canny)	Locates intensity discontinuities	Finds fruit outline and defect rim but edges are open, and specular highlights add false edges (Fig. 3d)	Moderate; gives shape, not area
Region-based (region growing, watershed)	Groups similar neighbouring pixels from seeds	Gives closed regions and areas; needs good seeds and is sensitive to texture	Good when seeded by colour

Proposed hybrid method

- **Step 1 – Colour space:** convert to Lab. The a* channel (green–red) separates a red apple from a grey belt almost independent of brightness.
- **Step 2 – Fruit mask:** Otsu on the smoothed a* channel, then morphological opening and contour extraction; watershed on the distance transform splits touching apples.
- **Step 3 – Highlight suppression:** pixels with very high L and low chroma are labelled specular and excluded.
- **Step 4 – Self-referenced defect detection:** inside each fruit, compare L with a heavily blurred ($\sigma \approx 12$) local reference of the same fruit; pixels darker than 0.45 × standard deviation + 14 grey levels are defect candidates. Because the fruit is compared with itself, brightness gradients cancel out.
- **Step 5 – Decision:** reject the apple if defect area exceeds 2 % of fruit area.

C. Demonstration with Tools

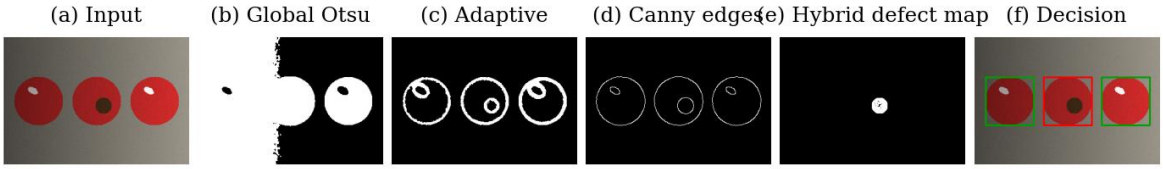


Figure 3. Three apples under a left-to-right brightness gradient with highlights; the middle apple is bruised.

Global Otsu (b) loses the darker left apple completely; adaptive thresholding (c) and Canny (d) outline the apples but do not give a reliable defect area. The hybrid method (e) marks only the bruise, and the decision stage (f) computed defect ratios of **0.0 %, 10.7 % and 0.0 %**, so only the middle apple is rejected (red box) while the two sound apples pass (green boxes), despite the highlight on the third and the shading on the first.

D. Innovation – Illumination-invariant, Self-referenced Inspection

- Using a chromatic channel for **where the fruit is** and a luminance channel relative to the fruit's own surface for **what is wrong with it** separates the two tasks that single-threshold methods mix.
- An explicit specular mask prevents shiny skin from being flagged as damage, and the area-ratio rule makes the decision interpretable to plant staff.
- Diffuse, enclosed lighting with a polariser would reduce the need for software highlight handling and is recommended as a hardware complement.

E. Challenges, Evaluation and Ethics

- Varieties with natural russet or striped skin may be flagged; per-variety thresholds should be calibrated.
- Evaluation: precision, recall and F1 of the *defective* class, plus the false-reject rate, because discarding good fruit has a direct cost.
- Rotating fruit shows only one side; two cameras or a roller are needed for full surface coverage.

F. Conclusion

No single classical technique is sufficient. A combination of Lab-colour fruit segmentation, specular masking and local-reference defect detection removed the effect of shadows, brightness gradient and reflection in the demonstration and identified the damaged apple correctly.

4. Lane-marking Detection for an Autonomous Vehicle

A. Problem Identification

The lane detector must output the lane boundaries in real time (≥ 25 frames per second) in sunlight, at dusk, in rain, and in the presence of tree shadows, faded paint and other vehicles. Failure is safety-critical, so the system must not only detect lines but also know **how confident** it is.

B. Technical Approach – Stages and their Challenges

Stage	Technique and purpose	Challenge	Counter-measure
1. Grayscale conversion	Reduces three channels to one for speed	Yellow paint and asphalt can have the same luminance, so yellow lanes disappear	Add HLS/HSV white and yellow colour masks
2. Filtering	Gaussian 5×5 removes noise and rain texture	Too much blur erases thin or distant lines	Kernel ≤ 5×5; median filter for rain droplets
3. Contrast enhancement	CLAHE equalises local contrast	Amplifies noise and wet-road glare	Moderate clip limit (≈ 2); apply to the lower half
4. Edge detection	Canny (thresholds 70/150) finds lane borders	Shadow boundaries, car edges and road cracks are also edges	Auto thresholds from median; orientation filter
5. Thresholding	Binary lane-candidate image by intensity or colour	Global threshold fails in shade or glare	Adaptive threshold or colour mask
6. Region of interest	Trapezoid mask keeps road area	Fixed mask fails on hills, curves and lane changes	Adapt ROI from the previous frame's vanishing point
7. Line fitting	Hough transform (rho 1, theta 1°, threshold 25, max gap 15)	Dashed or damaged markings break lines; vehicles create false lines	Large gap tolerance; slope filter; temporal smoothing

C. Demonstration with Tools

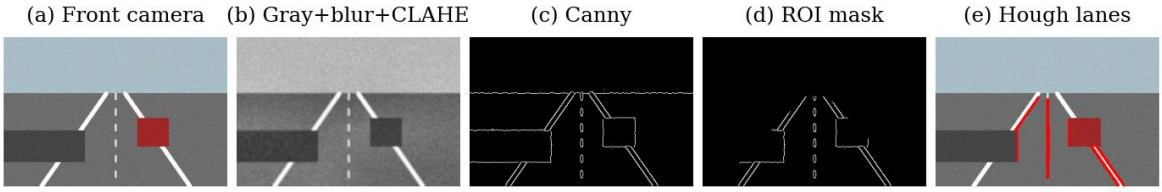


Figure 4. Lane pipeline on a synthetic road with a dark shadow patch and a vehicle.

The Canny result (c) contains the lane borders but also the horizontal edges of the shadow patch and the vehicle. The ROI mask (d) removes the sky and most of the left shadow, and Hough lines (e) lock on both lane edges. The run returned **13** candidate segments, which include a few false detections: the dashed centre-line and the shadow edge. This demonstrates the practical need for a geometric consistency check rather than trusting every line.

D. Innovation – Confidence-based Lane Tracking

- **Colour plus gray fusion:** a white mask (high lightness) and a yellow mask (hue 15–35°) are OR-ed with the edge map, recovering yellow lanes that grayscale loses.
- **Geometric validation:** accept a left/right line pair only if slopes have opposite signs, the lines intersect near a common vanishing point, and the lane width is plausible.
- **Temporal smoothing:** exponential or Kalman filtering of line parameters; if confidence is low (rain, glare) the last valid lane is held for a few frames and the driver-assistance module is notified.
- **Bird's-eye transform and polynomial fit** can replace Hough for curved roads.

E. Challenges, Evaluation and Ethics

- Rain reduces contrast and creates reflections; night adds headlight glare; damaged lines produce gaps. Classical methods degrade gracefully only if confidence is monitored.
- Evaluation: lateral position error in centimetres, detection rate per condition (clear, wet, night, shadow) and processing time per frame.
- Because errors may endanger people, the module must fail safe (alert the driver) and be validated on diverse data, including Indian road conditions with faint or missing markings.

F. Conclusion

Grayscale → blur → CLAHE → Canny/colour masks → ROI → Hough → temporal filter is a lightweight pipeline. Its limitations at each stage are known, and confidence-based tracking is the element that makes it robust enough for real driving conditions.

5. Waste Classification on a Conveyor Belt

A. Problem Identification

The camera must label each item as plastic, paper or metal while the objects overlap, change orientation, and cast shadows; metals and glossy plastics also produce strong reflections. Classification accuracy is limited less by the classifier than by the **quality of segmentation and the choice of features**, so the pipeline treats each stage as a source of invariance.

B. Technical Approach – Complete Pipeline

Stage	Techniques	Purpose / handling of real-world variation
1. Preprocessing	White-balance using a reference patch on the belt; median or NLM denoising; resize	Colour consistency across lamps; removes sensor noise
2. Enhancement	CLAHE on L/V channel; specular highlights detected (very bright, low saturation) and inpainted	Improves contrast of dull paper; stops reflections from corrupting colour features
3. Segmentation	Colour distance from the stored belt background, threshold, morphological opening; distance transform + watershed	Shadows have low chroma and are rejected; touching objects are separated
4. Feature extraction	Colour: HSV histogram. Texture: LBP, GLCM. Shape: Hu moments, solidity, aspect ratio. Reflectance: specular-pixel ratio	Rotation-invariant and illumination-tolerant descriptors
5. Classification	SVM (RBF) or Random Forest; CNN transfer learning (MobileNet) when data allow; reject option below a confidence limit	Maps feature vector to plastic / paper / metal / unknown

Stage	Techniques	Purpose / handling of real-world variation
6. Actuation and monitoring	Position from centroid triggers air-jet or arm; confusion matrix logged	Closes the loop and enables retraining

Handling real-world variation

Variation	Cause	Handling in the pipeline
Overlapping objects	Items piled on the belt	Distance-transform watershed; flag heavily occluded items for the next pass
Shadows	Overhead light or tall objects	Segment on chroma/colour distance, not brightness; diffuse lighting hood
Reflections	Metal foil, glossy plastic	Highlight mask and inpainting; keep the specular ratio as a *feature*
Noise	Low light, high conveyor speed	Median/NLM filter; short exposure with strobe light
Orientation	Random object pose	Hu moments, uniform LBP, rotation augmentation in training

C. Demonstration with Tools

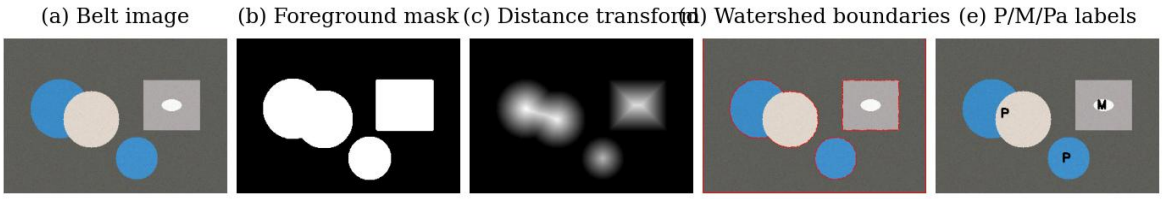


Figure 5. Synthetic belt scene: two plastic items (blue), one paper item (off-white) and a metal plate with a highlight.

The belt-distance foreground mask (b) ignores the grey belt and gives clean regions. The distance transform (c) shows one peak per object, and the watershed boundaries (d) delineate the isolated plastic item and the metal plate with its highlight. The overlapping plastic–paper pair is **not fully separated** in this run because its two

circles have similar radii and the neck between them is wide; the labels in (e) therefore treat it as one object. This is an honest limitation: it is solved by using h-maxima markers or a colour-aware watershed, and by flagging merged regions whose area is larger than any single item. The isolated plastic item is labelled P and the metal plate M, the latter because it has low saturation, lower brightness and a strong specular highlight.

D. Innovation – Physics-inspired Features and Reject Option

- **Specular ratio as a material cue:** metals have compact, intense highlights and low saturation, whereas paper is matte; using the reflection as information, rather than only removing it, helps the classifier separate metal from white paper.
- **Two-level decision:** a fast rule-based or SVM classifier handles clear cases, and uncertain items (confidence < 0.7) go to a CNN or to a manual-sorting bin, reducing contamination of recycled material.
- **Self-improving loop:** misclassified items logged during operation are added to the training set to adapt to seasonal waste changes.

E. Challenges, Evaluation and Ethics

- Dirty, wet, crushed and transparent items change colour and texture, and crumpled paper may resemble plastic bags.
- Evaluation: confusion matrix, per-class precision/recall/F1 and purity of each output bin, along with throughput (items per minute).
- Workers' safety and job transition around automated sorting should be considered, and the system should not capture identifiable people on the belt area.

F. Conclusion

A pipeline of preprocessing → highlight-aware enhancement → colour-distance segmentation with watershed → colour, texture, shape and reflectance features → confidence-aware classification addresses each real-world variation in a separate stage. The demonstration also shows where the design still needs refinement (overlapping items), which is what guides further engineering.